

UHD – MS Department – SPRING 2019
STAT 5309 –Design and Analysis of Experiments
CRN# 23403
Syllabus

I. Instructor

Name: HA NGUYEN, PhD

Office hours: Mon-Thurs 1:00-3:00PM

Office location: S-744 or N-615

E-mail: nguyenha@uhd.edu

II. Course

Name: STAT 5309- Design and Analysis of Experiments

Time: Thurs, 5:30-8:15PM. **Room:** One Main Building S-614

CRN : 23890

III. Objectives

Practical course in design and analysis of experiments for experiment workers and applied statisticians. R statistical software is used for analysis. This knowledge can be applied in many fields. Knowledge of basic statistics is required. Previous knowledge of R is not required but helpful. Basic and intermediate R will be introduced along the course.

IV. Textbook

Design and Analysis of Experiments by Montgomery (8th Edition)
Handouts

V. Supplemental readings

1. Design of Experiments, by R.O. Kuehl
2. [A First Course in Design and Analysis of Experiments](#), by Gary Oehlert
users.stat.umn.edu/~gary/book/fcdae.pdf
3. Design and Analysis of Experiments with R, by John Lawson
4. [An Introduction to R - R Project](#), by VN Venables
<https://cran.r-project.org/doc/manuals/r-release/R-intro.pdf>

VI. Assignments [subject to small changes]

There are R labs (35%), Midterm project and/or exam (25%) (which can be in class or take home), a Semester project (15%), Final exam (25%)[Take-home]. There are no quizzes.

VII. Topics/Chapters

- Overview and Basic Principles (Ch 1): Introduction to R. (Lab 0)
- Simple Comparative Experiments (Ch 2): randomization, paired and two-sample tests, introduction to R. (Lab 1)
- 1-Factor Factor Design (Ch 3) Completely Randomized Design (CRD). The Analysis of Variance : one-way ANOVA. Contrasts. Multiple comparisons tests. Assumptions checking. Fixed effects/ Random effects (Labs 2, 3).
- Completely Randomized Blocking Designs (CRBD). Latin Squares (Ch 4) (Labs 4A, 4B)
- Two-factor Factor Design (Ch 5) (Lab 5)
- 2-level (2^k) Factorial design (Ch 6) (Lab 6A)
- Fractional Factorial Design (Ch 8) (Lab 6B)
- Fitting regression models (Ch 10) (Labs 7)
- Response surface models (RSM) (Ch 11) (Lab 8)

Class lecture format: Lecture (1 hr), Break (15 mins); R (1 hr 15 mins)

VIII. UNIVERSITY POLICIES

ATTENDANCE - *It is important that you come to class!*

Fall 2017 UHD Attendance Policy: “Your failure to attend class (face to face or hybrid), engage course material (Online only); or make contact with faculty to adequately explain your absence by the 10th class calendar day of the semester will result in your being administratively dropped from this course. Being dropped from this course may affect your enrollment status and/or your financial aid eligibility.”

RESOURCE MATERIAL:

Center for Math and Statistics (C4MS or Math/Stat Tutors Lab) (925-N) (9th floor, North) (where they may receive assistance with understanding concepts, improving his skills, and working on homework problems. The Center is staffed with mathematics faculty and student assistants, and offers tutoring, use and/or checkout of course videotapes, use of calculators, use of computers with web access for math/stat homework and computer-aided math/stat drill, and use/loan of mathematics book resources on a walk-in basis. Hours: 8:00am-8:00pm M-Thu, 8:00am-2pm, Fri-Sat. Working exercises and discuss the material with others can greatly enhance your learning and overall achievement in the course. 3

STATEMENT ON REASONABLE ACCOMMODATIONS: The University of Houston-Downtown complies with Section 504 of the Rehabilitation Act of 1973 and the Americans with Disabilities Act of 1990, pertaining to the provision of reasonable academic adjustments/auxiliary aids for students with a disability. In accordance with Section 504 and ADA guidelines, UHD strives to provide reasonable academic adjustments/auxiliary aids to students who request and require them. If you believe that you have a disability requiring an academic adjustments/auxiliary aids, please contact the Office of Disability Services, One Main St. Suite 409-South Houston, TX 77002. (Office) 713-226-5227; (Website) www.uhd.edu/disability/; (Email) disabilityservices@uhd.edu.

IV. IMPORTANT DATES

- Jan 14, 2019 (Mon): Class starts
- Mar 11-16(Mon-Sat): SPRING BREAK
- Mar 28, 2019 (Thurs): Last day to withdraw with W
- Apr 29, 2019(Mon): Last day of instructions.
- Apr 30-May 1 (Tues-Wed): Reading days
- May 2-May 9 (Thurs-Thurs): Finals